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**Méthodes d'essai pour la mesure des propriétés physiques
des matières de balais pour machines électriques**

**Test procedures for determining physical properties
of brush materials for electrical machines**



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Révision de la présente publication

Le contenu technique des publications de la CEI est constamment revu par la Commission afin d'assurer qu'il reflète bien l'état actuel de la technique.

Les renseignements relatifs à ce travail de révision, à l'établissement des éditions révisées et aux mises à jour peuvent être obtenus auprès des Comités nationaux de la CEI et en consultant les documents ci-dessous:

- **Bulletin de la CEI**
Publié trimestriellement
- **Rapport d'activité de la CEI**
Publié annuellement
- **Catalogue des publications de la CEI**
Publié annuellement

Terminologie utilisée dans la présente publication

Seuls sont définis ici les termes spéciaux se rapportant à la présente publication.

En ce qui concerne la terminologie générale, le lecteur se reportera à la Publication 50 de la CEI: Vocabulaire Electrotechnique International (V.E.I.), qui est établie sous forme de chapitres séparés traitant chacun d'un sujet défini, l'index général étant publié séparément. Des détails complets sur le V.E.I. peuvent être obtenus sur demande.

Symboles graphiques et littéraux

Seuls les symboles graphiques et littéraux spéciaux sont inclus dans la présente publication.

Le recueil complet des symboles graphiques approuvés par la CEI fait l'objet de la Publication 117 de la CEI.

Les symboles littéraux et autres signes approuvés par la CEI font l'objet de la Publication 27 de la CEI.

Revision of this publication

The technical content of IEC publications is kept under constant review by the IEC, thus ensuring that the contents reflect current technology.

Information on the work of revision, the issue of revised editions and amendment sheets may be obtained from IEC National Committees and from the following IEC sources:

- **IEC Bulletin**
Published quarterly
- **Report on IEC Activities**
Published yearly
- **Catalogue of IEC Publications**
Published yearly

Terminology used in this publication

Only special terms required for the purpose of this publication are defined herein.

For general terminology, readers are referred to IEC Publication 50: International Electrotechnical Vocabulary (I.E.V.), which is issued in the form of separate chapters each dealing with a specific field, the General Index being published as a separate booklet. Full details of the I.E.V. will be supplied on request.

Graphical and letter symbols

Only special graphical and letter symbols are included in this publication.

The complete series of graphical symbols approved by the IEC is given in IEC Publication 117.

Letter symbols and other signs approved by the IEC are contained in IEC Publication 27.

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INTERNATIONAL ELECTROTECHNICAL COMMISSION

TEST PROCEDURES FOR DETERMINING PHYSICAL PROPERTIES OF BRUSH MATERIALS FOR ELECTRICAL MACHINES

FOREWORD

- 1) The formal decisions or agreements of the IEC on technical matters, prepared by Technical Committees on which all the National Committees having a special interest therein are represented, express, as nearly as possible, an international consensus of opinion on the subjects dealt with.
- 2) They have the form of recommendations for international use and they are accepted by the National Committees in that sense.
- 3) In order to promote international unification, the IEC expresses the wish that all National Committees should adopt the text of the IEC recommendation for their national rules in so far as national conditions will permit. Any divergence between the IEC recommendations and the corresponding national rules should, as far as possible, be clearly indicated in the latter.

PREFACE

This recommendation has been prepared by Sub-Committee 2F: Dimensions of carbon brushes, brush-holders, commutators and slip-rings, of IEC Technical Committee No. 2: Rotating Machinery.

A first draft was discussed at the meetings held in New Delhi in 1960, in Bucharest in 1962, in Aix-les-Bains in 1964, in Tokyo in 1965, in Baden-Baden in 1967, in London in 1968 and in Washington in 1970. As a result of this latter meeting, a revised draft was submitted to the National Committees for approval under the Six Months' Rule in February 1971.

The following countries voted explicitly in favour of publication:

Australia	Netherlands
Austria	Poland
Belgium	Portugal
Denmark	South Africa
Finland	Sweden
France	Switzerland
Germany	Turkey
Hungary	Union of Soviet
India	Socialist Republics
Israel	United Kingdom
Japan	United States
Korea (Democratic People's	of America
Republic of)	

TEST PROCEDURES FOR DETERMINING PHYSICAL PROPERTIES OF BRUSH MATERIALS FOR ELECTRICAL MACHINES

101. Scope

The properties determined by these tests are inherent in the brush materials themselves as distinct from performance characteristics in operation on electrical equipment.

Values of physical properties published in catalogues, etc., shall be based upon test methods, given in these recommendations.

The purpose of the different sections in these recommendations is to provide a uniform procedure for determining the following properties of brush materials:

- Density and porosity;
- Hardness;
- Resistivity;
- Transverse strength;
- Ash content.

Some test methods are suitable for use in production quality control, others only for more thorough investigations, using precise laboratory techniques.

It should be kept in mind that since brush materials are generally brittle, porous materials, it is reasonable that their properties vary much more than the same properties in metals.

Samples used for these tests must be representative of the batch of material involved using a recognized and/or agreed sampling procedure or technique.

102. Test specimen

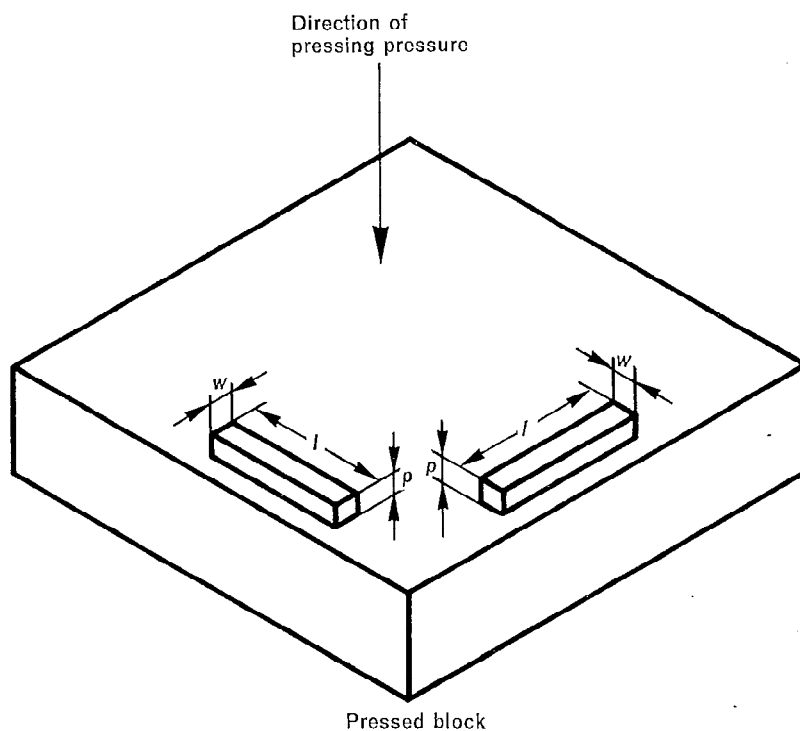
Some of the properties are to be determined on a standard test specimen (sample) with dimensions as given below. When the test specimen is cut from brush block material, it is necessary (in order to obtain a correct value when measuring the hardness and the transverse strength) that the dimensions (see figures 1 and 2) — thickness p , width w and length l — have a definite relation to the direction of pressing pressure or axis of extrusion. The length l must be perpendicular to the direction of compression for pressed blocks and must be along the axis of extrusion for extruded blocks. On brush material of laminated structure, the thickness p must be perpendicular to the lamination (parallel to the direction of compression).

The dimensions $p \times w \times l$ of the standard test specimen should be:

10 × 10 × 64 mm;
or
4 × 8 × 32 mm.

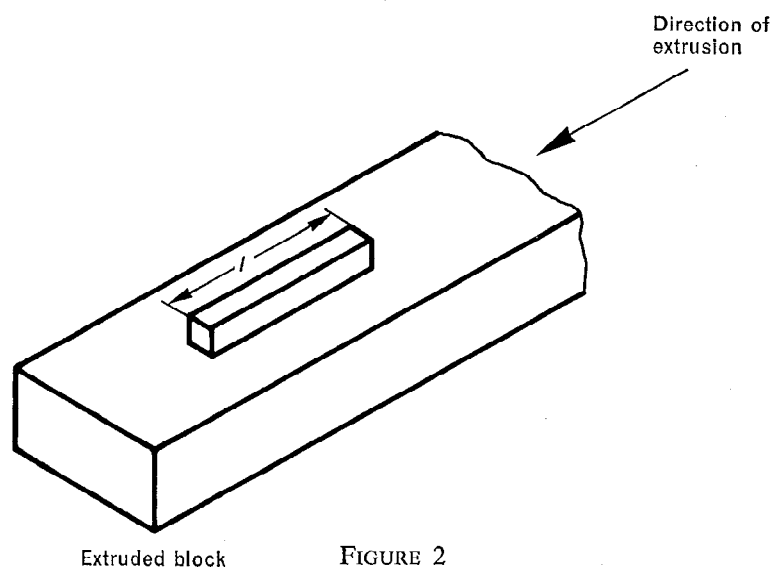
Where practicable, a test specimen may be cut from a manufactured brush.

*Relations between dimensions p , w and l of test specimen
and direction of pressing pressure or direction of extrusion*



Pressed block

FIGURE 1



Extruded block

FIGURE 2

In cases where it is not possible to cut a standard size test specimen from a manufactured brush, the size should be a matter of agreement between the user and the manufacturer. Caution should be used in testing smaller size pieces as the deviation may be great, due to difficulty in handling.

All adjacent surfaces must be at right angles. The specimen should be carefully machined to the surface finish normally used for carbon brushes. All traces of any die skin or extrusion skin should be removed.

The tolerance on the dimensions p and w is ± 0.04 mm and the tolerance on the dimension l is ± 0.8 mm.

When a large number of test specimens are concerned and density is one of the properties measured, it is more convenient to apply the tolerance of ± 0.04 mm to dimension l as well as to dimensions p and w .

In order to ensure squareness of the $p \times w$ section, the tolerance for the angles shall be $\pm 15'$. The parallelism defined by the difference between the dimensions of the two opposite sides at any cross-section perpendicular to the l dimension should be within 0.04 mm.

201. Density and porosity: general

Manufactured carbon is an inherently porous material and, therefore, both bulk (i.e. apparent) density and apparent porosity are useful to the manufacturer of carbon products as aids in control.

202. Nomenclature and definitions

The following letter symbols are used:

- m = mass of test specimen (dry) [g]
- V_b = bulk (apparent) volume of material (including accessible and closed pores) [cm³]
- V_o = volume of the accessible pores of the specimen [cm³]
- δ_b = bulk density [g/cm³]
- P_s = apparent porosity [percentage]
- d = diameter [mm]

The bulk density is then defined by the equation:

$$\delta_b = \frac{m}{V_b}.$$

The apparent porosity is defined by the formula:

$$P_s = \frac{V_o}{V_b} \cdot 100.$$

203. Methods for determining the bulk density and porosity

The bulk density δ_b may be determined by one of following standard methods:

- 204. Measurement-and-weight method;
- 205. Immersion method;
- 206. Oil-absorption method.

The porosity P_s may be determined by the oil-absorption method.

The measurement-and-weight method is suitable for production quality control and should preferably be used. The immersion method is considered to be an alternative and may be used in cases where the measurement-and-weight method cannot be applied or is unsuitable. The oil-absorption method is suitable for laboratory use.

All these methods are considered accurate to three significant figures, which is sufficiently accurate for checking these properties, since carbon products vary much more than metals.

Note. — The oil-absorption method may give unreliable values for material with a porosity higher than 30%.

204. Measurement-and-weight method**204.1 Equipment and test specimen**

Micrometer suitable for measuring the test specimen with an accuracy of ± 0.01 mm.

Scale or balance suitable for weighing the test specimen and having a sensitivity of 0.5% or better. (For specimens of the common small sizes, a balance having a capacity of 100 g and a sensitivity of ± 0.005 g is very suitable.)

Saw or other device for cutting the sample to a suitable size.

Grinding wheel or other device for machining the test specimen so that all adjacent surfaces are at right angles.

The specimen in accordance with Clause 102 or any rectangular or cylindrical sample of a suitable size the ends of which are plane and at right angles to the sides.

204.2 Test procedure

Using the micrometer, make the measurements necessary for calculating the volume of the test sample. This means measuring the thickness, width and length of rectangular pieces or the diameter and length of cylindrical pieces. These measurements are to be taken to the nearest hundredth of a millimetre.

Weigh the sample in grammes to three significant figures.

204.3 Calculations

Calculate the volume V_b of the sample and express in cm^3 :

$$V_b = p \times w \times l \quad \text{or} \quad \frac{\pi d^2}{4} \cdot l.$$

Calculate the bulk density δ_b in accordance with the following formula:

$$\delta_b = \frac{m}{V_b}.$$

205. Immersion method**205.1 Equipment and test specimen**

Container of water large enough to immerse the test sample.

Scale or balance in accordance with Sub-clause 204.1.

Grease or other waterproof masking material for coating the sample to prevent the absorption of water.

Thread to suspend sample while it is being weighed during immersion. The diameter of the thread shall not exceed 0.08 mm.

Test specimen. As this method may be used for irregularly-shaped samples, no preparation is necessary beyond making sure that the sample is clean, dry and free from extraneous metal or foreign material.

205.2 Test procedure

Weigh the sample in air. Coat as thinly as possible all surfaces of the sample with grease or other waterproof masking material. Weigh the sample in air again. The increase due to the coating should not exceed one part in 200. Express the result in grammes to three significant figures.

Suspend the sample by a piece of thread from the balance. Immerse it completely in water, taking care to remove all air bubbles, and weigh it while it is thus immersed. Use the minimum possible length of thread.

205.3 Calculations

When weighed in air, the force on the sample (due to gravity) is G_a . When the sample is immersed in water, the force on it (due to gravity) is also G_a , but, according to Archimedes' principle, another force $\gamma_w \cdot V_b$ (where γ_w is the specific weight of water and V_b is the volume of the sample) is also acting on the sample so that the weight measured in water is:

$$G_t = G_a - \gamma_w \cdot V_b$$

but

$$V_b = \frac{m}{\delta_b} = \frac{G_a}{g \cdot \delta_b}$$

and

$$\gamma_w = g \cdot \delta_w$$

where g is the acceleration due to gravity and δ_w is the density of the water. Thus:

$$G_t = G_a - \frac{\delta_w \cdot G_a}{\delta_b}$$

or

$$\delta_b = \delta_w \frac{G_a}{G_a - G_t}$$

The value of δ_w is very close to 1.

206. Oil-absorption method

206.1 Equipment and test specimen

Scale or balance in accordance with Sub-clause 204.1.

Hydrometer having a specific gravity range of 0.800 to 0.900 at 20 °C and graduated to 0.001.

Hydrometer cylinder.

Thermometer having a range of -10 °C to +100 °C.

Beaker having a wide mouth and a capacity of 1 000 cm³.

Vacuum desiccator or bell jar.

Vacuum pump capable of developing an absolute pressure of 20 mm of mercury (about 25 mb).

Water bath held at a temperature below 20 °C.

Petroleum oil grade SAE-10 (24-32 cSt at 50 °C) or kerosene.

Saw or other device for cutting the sample to a suitable size.

Grinding wheel or other device for smoothing off rough edges.

A test specimen with a weight of 25 to 100 g. If the sample is taken from a piece which originally had metal fittings, plating, etc., all such foreign material is to be removed. All surfaces must be ground to reasonable smoothness so that there are no burrs or sharp edges to chip off during the test. In general, metal-graphite grades and very dense materials should be tested in samples larger than those taken from other brush materials.

206.2 Test procedure

Weigh the sample in air to three significant figures for samples weighing up to 100 g, or to an accuracy of 1 in 200 or better for samples weighing more than 100 g.

Place the sample and a thermometer in the beaker, connect the pump to the beaker and reduce the pressure gradually to 20-60 mm of mercury (about 25-80 mb).

Without breaking the vacuum, gradually introduce oil or kerosene, preheated to 70 °C, into the beaker so that the sample is covered completely with oil.

Maintain the pressure at 20 to 60 mm of mercury (about 25-80 mb) until the sample stops emitting bubbles.

Cool the beaker and its contents to below 20 °C in the water bath. Remove the beaker from the bath and allow the temperature to rise to 20 °C.

At this point, remove the sample from the oil and wipe it dry.

Weigh the oil-impregnated sample in air to three significant figures or to 1 in 200 or better. Record this weight as G_2 .

Pour the oil from the beaker into the hydrometer cylinder, bring its temperature up to 20 °C and read its specific gravity. Record this value as δ_o .

Weigh the oil-impregnated sample submerged in water as described in Sub-clause 205.2. Record this weight as G_3 .

206.3 Calculations

The weight of the oil absorbed is $G_o = G_2 - G_a$. Thus the volume of oil absorbed is $V_o = \frac{G_o}{\delta_o}$. In analogy with the calculations in Sub-clause 205.3, the bulk density δ_b can be obtained as:

$$\delta_b = \delta_w \frac{G_a}{G_2 - G_3}$$

and the porosity

$$P_s = \frac{V_o}{V_b} \cdot 100.$$

301. Hardness: general

There are two different methods in general use for determining the hardness of brush materials:

302. Rebound method;

303. Indentation method.

It should be kept in mind that there is no general relation between the hardness of a brush material and the abrasiveness of the same material.

It is recognized that results obtained for a particular test method can be checked only against results obtained by the same method and standardized against equivalent standards.

The values of hardness measured on individual brushes may not be comparable to those taken on other brushes, or on brush blocks, unless the direction of pressing (or of extrusion) is the same.

302. Rebound method

This test method is suitable for production quality control, especially if the material to be tested is not very soft.

302.1 *Equipment and test specimen*

Scleroscope, SHORE model C2 or the equivalent, equipped with a special hammer for testing brush material.

Note. — The hammer and its components should be as follows:

- a) Mass: $2.2 \text{ g} \pm 5\%$;
- b) Outside diameter: $5.94 \begin{smallmatrix} + 0.013 \\ 0 \end{smallmatrix} \text{ mm}$ ($0.234 \begin{smallmatrix} + 0.0005 \\ 0 \end{smallmatrix} \text{ inch}$);
- c) Overall length: 20.7 to 21.4 mm (0.815 to 0.840 inch);
- d) Clearance between hammer and tube. At the top tube: 0.19 to 0.31 mm (0.0075 to 0.012 inch); at the bottom of the tube: 0.04 to 0.08 mm (0.0015 to 0.003 inch);
- e) Height from which hammer falls: $251.21 \begin{smallmatrix} + 0.13 \\ - 0.38 \end{smallmatrix} \text{ mm}$ ($9.890 \begin{smallmatrix} + 0.005 \\ - 0.015 \end{smallmatrix} \text{ inch}$);
- f) The tip of the hammer is to be a diamond, oval in shape and driven into a tapered hole in the hammer body. The body is to be water hardening drill-rod containing approximately 1.2% of carbon and is to be hardened and tempered after insertion of the diamond. Measured at the contact surface, the diamond is to have a curvature of $1.32 \text{ mm} \pm 5\%$ ($0.052 \text{ inch} \pm 5\%$), measured over a 0.61 mm (0.024 inch) long arc.

The test specimen may be of a size indicated in Clause 102, or any other convenient size. The piece to be tested should have the skin removed, and the surface should be machined smooth and free from defects, dirt, grit or other foreign material. A rectangular piece should have at least two parallel surfaces. A circular piece should have a uniform diameter, otherwise the hammer will strike the piece a glancing blow and false readings will result. Preferably, and if permissible, when testing circular pieces, two parallel flat surfaces should be machined to facilitate accurate testing.

302.2 *Setting up the instrument*

Place the instrument on a firm support.

Level up the instrument.

Check that the "plumb bob" on the front of the instrument can swing freely in the ring.

Check the calibration of the instrument using previously prepared reference samples.

Prepare the anvil for the test piece.

If the test piece is circular in cross-section and suitable parallel flat surfaces cannot be machined, use the vee block anvil. If the test piece is rectangular in cross-section, use the basic flat anvil (which is part of the instrument proper).

Note. — Keep the anvils clean. Wipe dust and dirt particles off the anvils with a soft lintless cloth. When a vee block anvil is used, wipe the flat anvil and the mating surface of the vee block anvil clean before placing them together. Dirt particles between the two anvils would cause false readings.

302.3 *Making the hardness test*

Place the piece on the appropriate anvil so that the hammer impact is applied in the direction of the pressing pressure in the case of brush block materials and perpendicular to the axis of extrusion for extruded materials.

Lower the tube against the test piece and hold it firmly while the test is being made.

Release the hammer, and read the height of the rebound.

If possible, measure the hardness at five points on each side of the test specimen.

To avoid random low readings giving a false result, disregard the lowest readings in each of the two groups of five readings. The average of the eight remaining readings is taken as being representative of the hardness of the specimen.

303. Indentation method

This test method is suitable for production quality control.

The test consists in forcing a ball into the surface of a test piece in two operations and measuring the permanent increase of the depth of indentation of this ball under specified conditions (see figure 3, page 23).

The hardness may be measured directly on manufactured brushes if:

1. The brush is not coppered and t , a and r (as defined in IEC Publication 136-1) are at least 10 mm; or
2. The brush is coppered and a and t are at least 10 mm and at the same time the part of r that is not coppered is at least 10 mm.

Measurements taken on a smaller specimen may be subject to error, due to possible effects of cracking, crushing and deformation.

For all other brushes, the measurement of the hardness is to be made on the brush block material before the brushes are cut to size.

303.1 Equipment and test specimen

Rockwell hardness test apparatus (penetrometer) with equipment for different loads F_0 , F_1 and F as given below. A device (such as a dashpot) should be incorporated to prevent the load from being applied suddenly: it should be adjusted so that, during the hardness test, the load increases from F_0 to $F = F_0 + F_1$ within 2 to 8 seconds (ISO Recommendation R 80).

Ball indenter made of polished material and having a diameter of D mm. No diameter of the ball should differ from the nominal diameter by more than the value corresponding to grade j6 of the ISO System. The ball should have a Vickers hardness of at least 850 (taking into account the curvature of the ball, when testing); it should be polished and free from surface defects. Any ball showing any deformation after the test greater than the tolerance specified above, or any surface defect, should be rejected and the corresponding test discarded.

Indicator (depth gauge) for measuring the permanent increase of depth of indentation. The indicator is to have a uniform dial scale reading from 0 to 130 where the distance between the scale divisions 30 and 130 should correspond to a depth of indentation of 0.200 mm, i.e. one scale unit corresponds to 0.002 mm. The indicator should be accurate to ± 0.001 mm.

Rigid support for the test piece. Test specimen shall be in accordance with Clause 102 or any sample of a suitable size with two parallel surfaces. Before the hardness test, the sample should be inspected to ensure that there are no cracks or other surface defects. The surface in contact with the support and the surface to be tested should be clean and free from foreign matter. When a hardness test is made on brush block material, it should be checked that the two parallel surfaces are carefully machined to the surface finish normally used for carbon brushes.

Note. — After each change or removal and replacement of the ball penetrator or the support, it should be ascertained that the new penetrator (or the new support) is correctly mounted in its housing.
Throughout the test, the apparatus should be protected from shock or vibration.

303.2 Test procedure

The test is carried out at ambient temperature.

It is important that the sample lies firmly on the support so that displacement cannot occur during the test. Unless otherwise agreed, the sample should be placed so that the direction of the load during the hardness test will be applied in the direction of pressing pressure in the case of brush block materials and perpendicular to the axis of extrusion for extruded materials.

Choose the position of any indentation in such a way that the distance between the centres of two adjacent indentations or from the centre of any indentation to the edge of the sample or to any cavity will be at least one ball diameter or 2.5 times the diameter of the impression diameter, unless otherwise agreed.

Choose a combination of ball diameter D , preliminary load F_0 and additional load F_1 as follows:

Combination 1: for natural graphite materials;

Combination 2: for metal graphite materials;

Combinations 3, 4 and 5: for metal graphite, electrographite and hard carbon materials.

Combinations number	1	(2) *	3	4	5	
D	10	(10) *	10	10	10	mm
F_0	10	(10) *	10	10	10	kgf
F_1	10	(30) *	50	90	140	kgf
$F = F_0 + F_1$	20	(40) *	60	100	150	kgf

* These combinations should be avoided as far as possible.

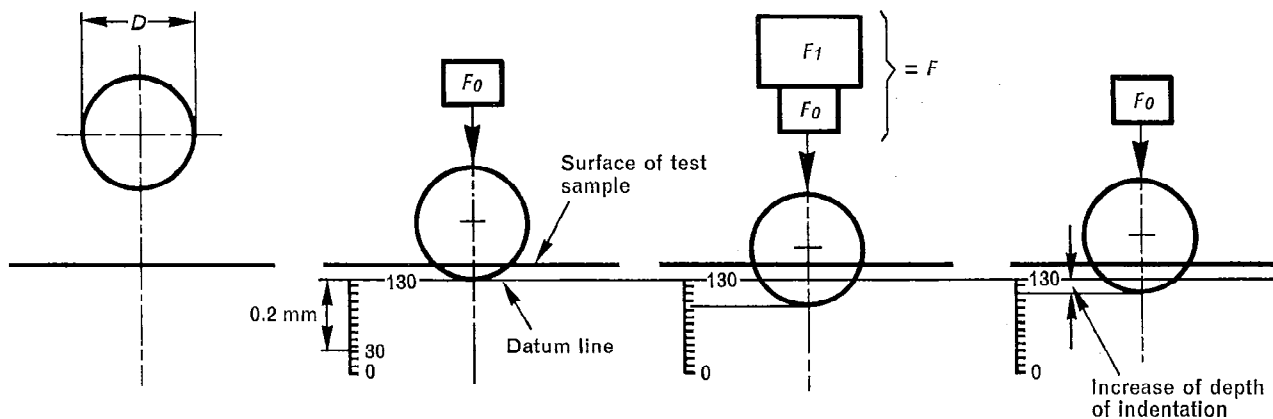
In accordance with the practice laid down by the International Organization for Standardization (ISO), the loads for all hardness testing are expressed in kgf. This decision was taken as it is considered impracticable to change the loads on all standard hardness testing machines to daN.

Where none of the listed combinations gives a suitable reading, a different size ball may be used, and this information included in the report. In this situation, preference should be given to using a 5 mm diameter ball, with the combinations of load given in the table.

Place the ball against the surface of the test piece. Apply pressure gradually to the ball perpendicular to the surface without sudden shock until the preliminary load F_0 of 10 ± 0.2 kgf is attained.

Set the dial of the indicator (depth gauge) at the initial position.

Increase the load, without sudden shock and within 2 to 8 seconds, by the value of additional load F_1 , thus obtaining a total load $F = F_0 + F_1$. The tolerance on F_1 is $\pm 0.5\%$.



Hardness test: indentation method

D = diameter of ball penetrator

F_0 = preliminary load

F_1 = additional load

FIGURE 3

When the needle of the indicator (depth gauge) is steady, which generally takes up to 10 seconds according to the material being tested, remove the additional load F_1 so as to bring the load back to the preliminary load F_0 .

In special cases, where the brush material under these conditions of the test, shows greater time dependent plasticity, F_1 is removed 20 to 25 seconds after the indicator commences to move.

Read from the dial the permanent increase of depth of indentation while the preliminary load F_0 is still applied. Deduce, if necessary, the corresponding Rockwell hardness number. (Most dial scales give a direct reading of the Rockwell hardness number on the red scale.) If, when using one combination of the table, the average value obtained is greater than 110 or less than 30, the test should be repeated using a more suitable combination.

Make the following number of hardness tests on each sample:

- On manufactured brushes where the two dimensions of the surface to be tested are both less than 16 mm:
 - 1 test on each of the two surfaces;
- On manufactured brushes where the two dimensions of the surfaces to be tested are both at least 16 mm:
 - 2 tests on each of the two surfaces;
- On brush block material:
 - 5 tests on each of the two surfaces.

Record, as the hardness, the arithmetical mean value of the readings taken on each sample. However, in the case of brush block material it is advisable to disregard the lowest value obtained on each face; the arithmetical mean of the eight remaining results is then considered representative of the hardness.

The hardness value obtained for a specified load should be designated by the letters *HR* in combination with an index figure stating which ball diameter D and total load F had been used, for instance $HR_{10/60}$.

401. Resistivity: general

There are several different methods in general use for determining the resistivity of brush materials. The methods used most frequently for production quality control are the following:

- 402. Voltmeter-ammeter method;
- 403. Kelvin bridge method.

402. Voltmeter-ammeter method

402.1 Equipment and test specimen

Source of direct current providing a range of 0 to 6 volts and 40 A (for instance storage battery and variable resistance).

- Millivoltmeter for direct current having a resistance of at least 1 000 Ω/V and preferably having the following scales:

- 0 to 300 mV
- 0 to 150 mV
- 0 to 75 mV
- 0 to 30 mV
- 0 to 15 mV
- 0 to 7.5 mV
- 0 to 3 mV

— Ammeter for direct current preferably having the following scales:

- 0 to 75 A
- 0 to 30 A
- 0 to 15 A
- 0 to 7.5 A
- 0 to 3 A

(All instruments should be in accordance with class 1.5 of IEC Publication 51, Direct Acting Indicating Electrical Measuring Instruments and Their Accessories.)

- Switch for breaking and closing the test circuit.
- Two current contacts comprising resilient (for instance copper gauze) or self-aligning contact surfaces. The two surfaces are to be suitably arranged to hold between them a standard test specimen with dimensions in accordance with Clause 102.
- Sharp potential pointers of stainless steel having a spacing not less than 30% and not greater than 50% of the length of the specimen. The tolerance on the spacing is $\pm 1\%$ of its nominal value.
- Test specimen in accordance to Clause 102.

402.2 Test procedure

The test specimen should be tested at a temperature of $25 \pm 10^\circ\text{C}$.

Measure and record the dimensions of the test specimen to an accuracy of 0.5% or better.

Put the test specimen lengthwise between the two current contacts.

Press the potential pointers against one of the surfaces $w \times l$ of the specimen. The force on the potential pointers is considered to be sufficient if the needle of the millivoltmeter is steady.

Check that the potential pointers are symmetrically located in relation to the axis of the surface $w \times l$.

Record the potential pointer spacing.

Note. — This potential pointer spacing is to be checked at regular intervals. The check is made by pressing the pointers against a flat ground surface of a piece of brush material on which clear, well defined marks are easily obtained. The distance between the centres of these marks is then measured by means of a microscope.

Pass the current through the test circuit. Choose suitable scale on ammeter and millivoltmeter and adjust the value of the current, by means of the variable resistance, so that readings of over 50% full scale deflection are obtained on both instruments.

Notes 1. — Suggested values of current, potential pointer spacing and millivoltmeter scale are given in Table I.

- 2. — The readings should be taken quickly in order to avoid heating the specimen since this may influence the value of the resistivity.

Record the ammeter and millivoltmeter readings and break the current through the test circuit.

Disengage the potential pointers from the surface of the test specimen.

Remove the pressure from the two current contacts, turn the test specimen over, so that the potential pointers are in contact with the other surface $w \times l$, and apply the same force as before.

Repeat the test procedure using the same current.

The arithmetical mean value of the voltage drops measured on the two surfaces $w \times l$ is used for calculation of the resistivity.

402.3 *Calculations*

The resistivity (specific resistance) ρ is calculated in accordance with the following formula:

$$\rho = \frac{U \cdot p \cdot w}{I \cdot l_u}$$

where:

ρ = resistivity (specific resistance) of the brush material, in ohm-metres

U = voltage drop between the potential pointers, in volts

p = thickness of test specimen, in metres

w = width of test specimen, in metres

I = current through test specimen, in amperes

l_u = distance between the potential pointers, in metres.

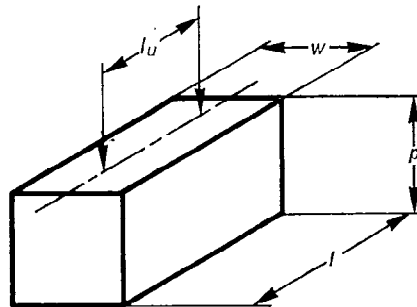


FIGURE 4

Note. — It is recommended that the resistivity of brush materials should be quoted in microhm-metre units ($\mu\Omega\text{m}$). ρ is obtained in microhm-metre if, in the formula above, U is measured in millivolts, I in amperes and p , w and l_u all in millimetres.

403. **Kelvin bridge method**403.1 *Equipment and test specimen*

Source of direct current, ammeter, resilient (for instance copper gauze) or self-aligning current contacts and potential pointers in accordance with Sub-clause 402.1.

Kelvin bridge having a range of 10 microhms to 1 ohm.

Galvanometer sensitive to 1 microampere per scale division.

Switch for passing the current intermittently through the specimen.

Test specimen in accordance with Clause 102.

403.2 *Test procedure*

The specimen should be tested at a temperature of $25 \pm 10^\circ\text{C}$.

Measure and record the dimensions of the test specimen to an accuracy of 0.5% or better.

Place the test specimen lengthwise between the two current contacts.

Connect the test specimen to the Kelvin bridge correctly and carefully.

Press the potential pointers against one of the surfaces $w \times l$ of the specimen. The force on the potential pointers is considered to be sufficient if the needle of the millivoltmeter is steady.

Check that the potential pointers are symmetrically located in relation to the axis of the surface $w \times l$.

Record the potential pointer spacing.

Note. — The potential pointer spacing is to be checked at regular intervals. This check is made by pressing the pointers against a flat, ground surface of a piece of brush material on which clear well-defined marks are easily obtained. The distance between the centres of these marks is then measured by means of a microscope.

Pass current intermittently, by means of the switch, through the test circuit. When the current is on, adjust the bridge resistance successively until the galvanometer has zero deflection. Read the resistance R of the specimen between the potential pointers.

Note. — The adjustment of the bridge resistance should be made as quickly as possible in order to avoid any heating of the specimen that may influence the value of the resistance.

Disengage the potential pointers from the surface of the test specimen.

Remove the pressure from the two current contacts, turn the test specimen over, so that the potential pointers are in contact with the other surface $w \times l$ of the specimen, and apply the same force as before.

Repeat the test procedure using the same current.

The arithmetical mean value of the resistances measured on the two surfaces $w \times l$ is used for calculation of the resistivity.

403.3 Calculation

The resistivity ρ is calculated in accordance with the following formula:

$$\rho = \frac{R \cdot p \cdot w}{l_u}$$

where:

ρ = resistivity (specific resistance) of the brush material in ohm-metres

R = resistance of test specimen, between potential pointers, in ohms

p = thickness of test specimen, in metres

w = width of test specimen, in metres

l_u = distance between potential pointers, in metres.

Note. — ρ is obtained in microhm-metres if, in the formula above, R is measure in milliohms and p , w and l_u all in millimetres.

It is recommended that the resistivity of brush materials should be quoted in microhm-metre units ($\mu\Omega m$).

TABLE I
Suitable test conditions

Test specimen	l_u	ρ max	I	U max
$p \times w \times l$ mm mm mm	mm	$\mu\Omega m$	A	mV
10 × 10 × 64	25	600.00	2.00	300.0
		300.00	4.00	300.0
		150.00	8.00	300.0
		75.00	8.00	150.0
		37.50	8.00	75.0
		15.00	8.00	30.0
		7.50	8.00	15.0
		3.75	8.00	7.5
		1.50	20.00	7.5
		0.75	40.00	7.5
		0.60	20.00	3.0
		0.30	40.00	3.0
		600.00	1.00	300.0
		300.00	2.00	300.0
4 × 8 × 32	16	150.00	4.00	300.0
		75.00	4.00	150.0
		37.50	4.00	75.0
		15.00	4.00	30.0
		7.50	4.00	15.0
		3.75	4.00	7.5
		1.50	10.00	7.5
		0.75	20.00	7.5
		0.60	10.00	3.0
		0.30	20.00	3.0
		600.00	1.00	300.0
		300.00	2.00	300.0
		150.00	4.00	300.0
		75.00	4.00	150.0

501. Transverse strength

501.1 Equipment and test specimen

Pressure device with a suitable pressure gauge or scale (for example, any hydraulic or mechanical press). The device should be capable of developing and measuring forces up to about 1 500 N with an accuracy of $\pm 2\%$.

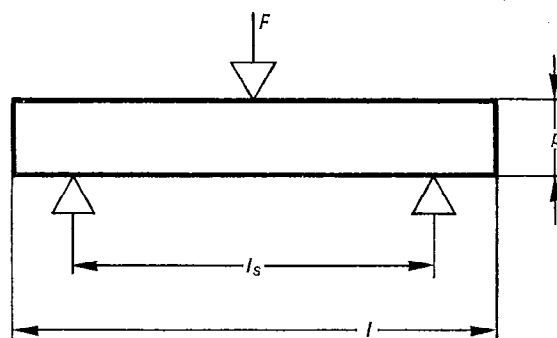


FIGURE 5

The three knife edges are rounded:

The radius for the loading knife edge should be: $r = 4$ to 5 mm;

The radius for the supporting knife edges should: $r = 1 \text{ mm} \pm 0.5 \text{ mm}$.

The outer two knife edges (the supporting knife edges) should lie in the same plane and be parallel and the distance between their centres should not differ from the nominal value by more than ± 0.2 mm. The knife edge in the middle (the loading knife edge) should always be parallel to the two outer ones and should be movable in a plane perpendicular to the plane through the two supporting knife edges and located midway between them.

The test specimen should be in accordance with Clause 102.

501.2 Test procedure

Check that the distance l_s between the two supporting knife edges is equal to the value recommended in Table 2. Record the distance l_s .

Check the dimensions of the test specimen and record them. w and p should be measured to an accuracy of 0.5% or better.

Place the specimen symmetrically on the two supporting knife edges with the dimensions l perpendicular to the knife edges and the dimension w parallel to them.

Apply, by means of the loading knife edge, the stress at the centre of the specimen, midway between the two supporting knife edges.

Notes 1. — The loading knife edge should be able to move in its vertical plane and parallel to the supporting knife edges without causing any measurable parasitic friction.

2. — The force should be applied at a uniform rate without shock so that the rupture takes place in not less than 5 seconds.

Record the total force net F at which the test specimen ruptures.

501.3 Calculations

The transverse strength σ is calculated in accordance with the following formula:

$$\sigma = \frac{3}{2} \times \frac{l_s}{w \cdot p^2} \times F$$

where:

σ = transverse strength of the brush material, in MN/m²

F = total net force at the instant of rupture, in N

l_s = distance between supporting knife edges, in mm

w = width of test specimen, in mm

p = thickness of test specimen, in mm.

Table II below gives, for each of the two standard test specimens, the recommended value for l_s and the corresponding value for the coefficient c .

$$c = \frac{3}{2} \times \frac{l_s}{w \cdot p^2}$$

σ in MN/m² is then simply obtained from the formula:

$$\sigma = c \cdot F.$$

TABLE II

Dimensions of standard test specimen			Recommended values of l_s mm	Corresponding value of c $\frac{1}{\text{mm}^2}$
p mm	w mm	l mm		
10	10	64	50.0	0.075
4	8	32	25.6	0.300

Whenever possible, test specimens of 10×10×64 mm should be used. If, for any reason, the test specimen of 4×8×32 mm has to be used, this should be recorded.

601. Ash content: general

The materials used for electrical brushes may contain relatively small amounts of other chemical elements and compounds. These may occur normally, as in some grades of natural graphite, or may purposely be added, as for example silicon carbide. These materials are grouped under the general category of "ash". They may have some influence on the performance in assisting film formation or provide some "scrubbing" action. It is therefore desirable to have a test method for the determination of ash content. It is used primarily by the manufacturer for production control.

601.1 Equipment

Furnace for temperature up to 1 000 °C.

Equipment for regulating the temperature in the furnace.

Equipment for measuring the temperature up to 1 000 °C.

Drying oven for temperature up to at least 110 °C.

Thermometer for measuring temperature up to 110 °C.

Desiccator with a suitable hygroscopic substance, for instance silica gel. The hygroscopic substance is to be renewed when necessary.

Note. — The drying oven, the thermometer and the desiccator are not necessary if other precautions have been taken.

Shallow containers of a suitable size and material. They should be marked for identifications and heated to 1 000 °C until their weight is stabilized. When not being used they shall be kept in the drying oven.

Tongs for moving the containers.

Analytical balance capable of weighing one container plus about 5 g of brush material with an accuracy of ± 0.2 mg.

Device for obtaining a sample of the material, e.g. saw or drill.

Spatula, pestle and mortar.

601.2 *Test specimen*

Not more than about 5 g of brush material should be taken from manufactured brushes or brush block material which should be representative of the lot to be tested (1 or 2 g is usual for a sample taken from a brush).

The pieces should be granulated in a mortar until all particles are about 0.5 mm or less in all dimensions.

Note. — Take every precaution in the preparation of sample to exclude every form of foreign matter.

601.3 *Test procedure for non-metallic brushes*

Dry the test specimen in an oven at 100 ± 10 °C to constant mass (this may take up to 2 hours).

Weigh a clean, dry container to within 0.2 mg.

Record the weight as G_1 .

Add 1 to 5 g of dried sample and weigh the container plus sample to within 0.2 mg.

Record the mass as G_2 .

Put the container and sample in the furnace and increase the temperature slowly to at least 800 °C, but not above 1 000 °C.

During this period, note and record any recognizable odours (for instance phenol or sulphur).

Continue to burn sample until no evidence of carbon remains.

Note. — If all requirements mentioned above are fulfilled, 12 hours heating are sufficient for all materials to ensure complete ashing to constant mass.

Remove the container with its contents from the furnace. Cover the container while cooling until it reaches about 200 °C and then transfer it to a desiccator. Do not weigh until it reaches the ambient temperature in the balance room.

Weigh the container with its contents to within 0.2 mg.

Record this mass as G_3 .

601.4 *Calculation*

The ash content is calculated in accordance with the following formula:

$$a = 100 \frac{\text{mass of residue}}{\text{mass of original dry sample}} \% = 100 \frac{G_3 - G_1}{G_2 - G_1} \%$$

Unless otherwise agreed between user and manufacturer, at least 5 tests are to be made in accordance with the specifications given above.

The arithmetical mean value of the results should be calculated and considered as the true ash content; the range should be quoted.

If, at the time the order was placed, the manufacturer specified a value of the ash content, the true ash content should be compared with the specified value and should fall between the limits shown on the diagram in Figure 6, page 41.

Note. For the time being, the limits given in Figure 6 are for guidance only.

601.5 *Test procedures for metallic brushes*

Because different methods of test for ash content in metal graphite brushes would be required depending on the presence or absence of major elements or compounds such as silver, tin, molybdenum (MoS_2), plastics, etc., no test method is presented here.

FIGURE 6

Upper and lower limits of true ash content for different values of specified ash content

